

University of Victoria
Department of Mechanical Engineering



**University
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Final Project Report
Application of Integrated CAD for Structural
Optimization of a Near-Shore Smart Buoy

MECH 410 - Computer-Aided Design and Engineering
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1.0 Introduction

1.1 Project Background

Ocean monitoring buoys are used to collect environmental data such as wave height, wave period, wind speed, and water temperature. These buoys are commonly deployed in near-shore environments where they are exposed to waves, currents, and mooring forces. Structural integrity is critical because failure of the buoy body could result in loss of instrumentation and data.

This project focuses on the structural analysis of a near-shore smart buoy hull that is currently being developed as part of a separate capstone project. Due to manufacturing constraints, the buoy prototype is produced using additive manufacturing. The buoy body is printed in multiple sections which are adhesively bonded together to form the final geometry. Because the buoy experiences loading from mooring tension and environmental forces, it is important to evaluate the structural integrity of the hull and the bonded joints.

The goal of this project is to apply an integrated CAD and CAE workflow to evaluate structural performance under realistic loading conditions. Finite Element Analysis (FEA) is used to determine stress and displacement in the buoy structure. The analysis also considers the differences between a prototype PLA buoy and a potential production version manufactured using polyethylene.



Figure 1. Near-shore smart buoy

1.2 Problem Definition

The buoy structure must withstand loading caused by mooring tension and environmental forces such as waves and currents. Because the prototype buoy is manufactured using additive manufacturing, the structure is divided into multiple sections that are bonded together using adhesive. These bonded interfaces introduce potential stress concentration regions that may affect structural performance.

The primary structural loading condition occurs at the bottom of the buoy where the mooring line is attached. The mooring line applies tension to the hull as environmental forces act on the buoy. The magnitude of the mooring tension depends on wave height, wave period, and current velocity at the deployment location.

The buoy walls must be thick enough to prevent structural failure but thin enough to minimize material usage and cost. The prototype buoy is manufactured using PLA, while a production version of the buoy could be manufactured using polyethylene. Because the material properties of PLA and polyethylene differ significantly, structural performance must be evaluated for both materials.

The main engineering problem is to determine whether the buoy hull is structurally safe under expected loading conditions and whether the wall thickness can be reduced while maintaining structural integrity.

1.3 Objectives of the Study

The objectives of this project are:

- Develop a CAD model of the buoy geometry for structural analysis
- Determine realistic loading conditions using hydrostatic simulation
- Perform Finite Element Analysis on the buoy hull using PLA material properties
- Evaluate stress at the bonded joints between printed sections
- Perform Finite Element Analysis using polyethylene material properties for comparison
- Apply structural optimization to reduce wall thickness while maintaining structural integrity
- Identify opportunities to reduce material usage and manufacturing cost

2.0 Scope of the Project

2.1 Buoy Geometry

The buoy geometry consists of a hollow cylindrical body with a domed upper surface. The outer diameter of the buoy is 18 inches and the wall thickness is modeled as 0.5 inches (exactly what the prototype is designed at). The bottom portion of the buoy contains a reinforced region where the mooring line is attached. The buoy geometry was created using Siemens NX CAD software.

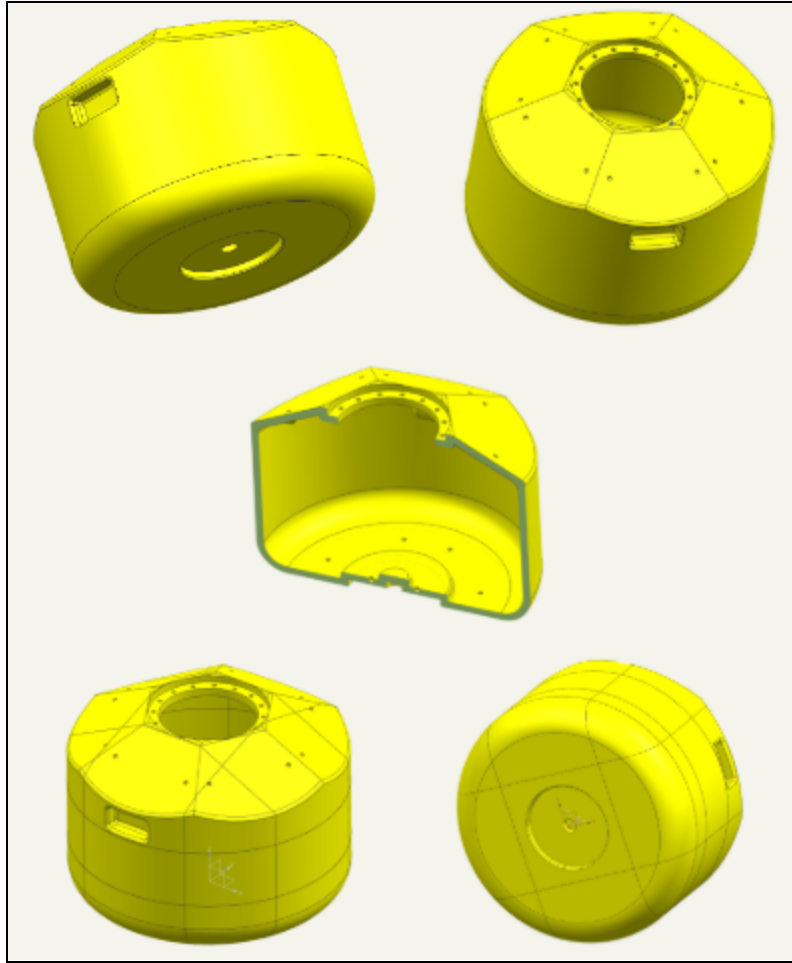


Figure 2. CAD model of buoy geometry

2.2 Segmented Manufacturing Approach (Prototype)

The buoy prototype is manufactured using additive manufacturing due to size limitations of the available 3D printer. Because the buoy is larger than the printer build volume, the structure must be divided into multiple sections that can be printed separately and then bonded together using adhesive.

The segmented manufacturing approach introduces bonded interfaces that may experience stress concentrations when the buoy is subjected to loading. These interfaces are important regions for structural evaluation because adhesive bonds may fail if stresses exceed allowable limits.

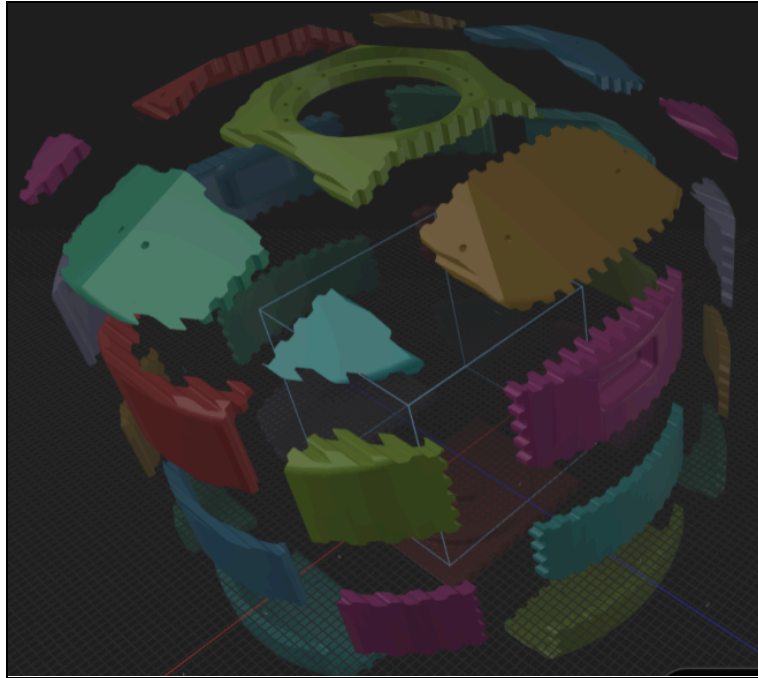


Figure 3. Segmented buoy geometry for additive manufacturing

2.3 Realistic Manufacturing Approach (Polyethylene)

While the prototype buoy is manufactured using PLA, a production version of the buoy would be manufactured using polyethylene (PE) due to its improved durability and resistance to environmental conditions. PE is commonly used in marine applications because of its impact resistance, corrosion resistance, and long-term performance in water environments.

To evaluate the structural feasibility of a production design, the buoy geometry was also analyzed using PE material properties. Comparing PLA and PE allows for evaluation of how material selection influences stress distribution and structural performance.

2.4 Loading Conditions and Environmental Assumptions

The buoy is subjected to loading caused by mooring tension and environmental forces. The primary structural load occurs at the bottom of the buoy where the mooring line is attached. The magnitude of the mooring force depends on environmental conditions including wave height, wave period, wind speed, and current velocity.

Hydrodynamic simulation software was used to estimate the maximum mooring tension under expected near-shore conditions. These conditions represent severe ocean environments to ensure conservative structural design.

In addition to mooring tension, the buoy experiences hydrostatic pressure caused by water surrounding the submerged portion of the hull. hydrostatic drag forces also act on the submerged portion of the buoy due to ocean currents.

The determination of mooring tension using hydrostatic simulation is discussed further in section 3.1.

2.5 Performance Metrics

Structural performance of the buoy was evaluated using the following metrics:

- Maximum von Mises stress
- Maximum displacement
- Stress distribution near bonded joints
- Structural response under hydrostatic loading
- Effect of material selection on structural performance

These performance metrics were used to determine whether the buoy structure satisfies strength requirements and whether wall thickness can be reduced through optimization.

3. CAD/CAE Modelling Tool Selection

3.1 ProteusDS for Mooring Load Simulation

ProteusDS hydrodynamic simulation software was used to estimate the mooring tension acting on the buoy under realistic ocean conditions. The buoy is intended for deployment in near-shore environments along Vancouver Island, including locations such as Arbutus Cove, Sombrio Beach, and Jordan River. These locations experience significant wave activity during Pacific winter storm events.

The buoy is being designed by the capstone team to withstand Sea State 7 conditions according to the Douglas Sea Scale. Sea State 7 corresponds to very rough ocean conditions with typical wave heights between 6 m and 9 m. Designing for Sea State 7 ensures the buoy structure can survive severe environmental loading while maintaining functionality.

Because the buoy is deployed in near-shore environments approximately 100 m to 300 m from the shoreline, wave characteristics at the deployment distance are reduced compared to fully developed offshore wave conditions. Wave energy decreases as waves interact with the seabed and coastal bathymetry. Therefore, representative near-shore environmental parameters were estimated based on Sea State 7 conditions adjusted for reduced fetch distance.

The following environmental parameters were used as inputs for the ProteusDS simulation:

- Current velocity: 0.5 m/s
- Wave height: 6 m
- Wave period: 15 s
- Wind speed: 40 km/h

These values represent conservative near-shore storm conditions that are expected to produce significant loading on the buoy structure.

ProteusDS simulations were performed by modeling the buoy as a rigid floating body connected to a fixed seabed anchor using a mooring line. Wave, current, and wind inputs were applied simultaneously in order to simulate realistic environmental loading conditions. The simulation outputs included buoy motion response and tension forces in the mooring line.

The maximum mooring tension obtained from the simulation was approximately 340 N. This force represents the primary structural loading condition applied to the buoy hull. The force was applied at the mooring attachment point on the bottom surface of the buoy in the FEA model.

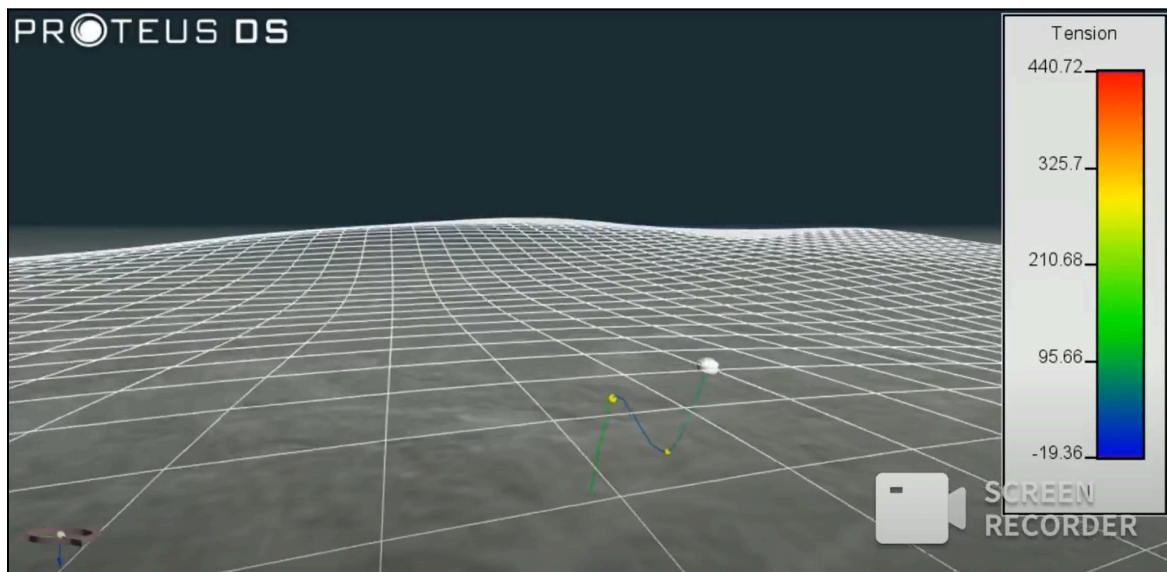


Figure 4. ProteusDS model showing mooring configuration and predicted tension forces for Sea State 7 near-shore conditions

3.2 Hydrostatic Force

The buoy experiences hydrostatic pressure on the submerged portion of its exterior surface. Hydrostatic pressure results from the weight of the water surrounding the buoy and increases with depth below the waterline. In Siemens NX, hydrostatic pressure can be applied directly as a function of depth, allowing the software to calculate the pressure distribution automatically.

In order to apply hydrostatic pressure in NX, the location of the waterline must first be determined. The waterline corresponds to the draft depth, which represents how deep the buoy sinks into the water when floating at equilibrium.

The draft depth was calculated using the buoy mass and displaced water volume. At equilibrium, the buoy weight is equal to the buoyant force. The buoyant force is equal to the weight of the displaced water.

The displaced volume can be calculated as:

$$V_{displaced} = \frac{m_{buoy}}{\rho_{water}}$$

Because the buoy has a circular cross section, the submerged volume can also be expressed as:

$$V_{submerged} = \pi r^2 \delta$$

Solving for draft (δ):

$$\delta = \frac{m_{buoy}}{\rho_{water} \pi r^2}$$

Using the buoy properties:

- $m_{buoy} = 10.488kg$
- $\rho_{water} = 1025kg/m^3$
- $r = 0.2285m$

$$\delta = 2.46in$$

The draft represents the vertical distance from the bottom of the buoy to the waterline. This defines the portion of the buoy surface exposed to hydrostatic pressure.

NX automatically applies the varying pressure distribution once the draft depth and fluid density are defined.

3.3 Granta EduPack Material Database for PLA and PE Properties

Material properties for PLA were obtained using the Ansys Granta EduPack material database. Granta EduPack provides experimentally measured material properties that can be used for engineering simulations.

Relevant material properties for PLA include:

- Young's modulus
- Poisson's ratio
- Density
- Tensile strength
- Thermal properties

These properties were used to create a custom material definition in Siemens NX. The full material property table used in the simulation is provided in Appendix B.

3.4 Siemens NX for CAD Modelling and FEA Simulation

Siemens NX was used to create the CAD geometry and perform Finite Element Analysis of the buoy structure. NX allows integration of CAD modelling and CAE simulation within a single software environment.

The buoy geometry was recreated in NX where material properties, boundary conditions, and loading conditions were defined. A tetrahedral mesh was generated to represent the geometry for structural analysis.

Boundary conditions were applied to represent the connection between the mooring line and the buoy structure. Forces obtained from hydrodynamic simulation were applied to evaluate stress distribution and structural response.

NX optimization tools were also used to evaluate the effect of reducing wall thickness on structural performance for both PLA and polyethylene material models.

3.5 Justification for Integrated CAD workflow

An integrated CAD and CAE workflow was used to ensure consistency between geometry, material properties, and structural simulations. The buoy was modeled and analyzed within

Siemens NX, allowing design changes such as wall thickness adjustments and material substitutions to be updated directly in the simulation environment.

This approach allowed efficient comparison between the PLA prototype and the polyethylene production model while maintaining identical geometry and loading conditions. Material properties from Granta EduPack and mooring loads from ProteusDS were applied directly within NX, improving accuracy and traceability.

Overall, the integrated workflow reduced modelling errors, improved efficiency during optimization, and ensured consistent evaluation of structural performance for both material cases.

4.0 Technical Challenges and Special Features

4.1 Simulation of Mooring Line Tension from Ocean Conditions

A key technical challenge in this project was estimating the mooring line tension that the buoy may experience under realistic ocean conditions. The magnitude of the mooring force depends on environmental parameters such as wave height, wave period, wind speed, and current velocity. These parameters can vary significantly depending on the deployment location, time of year, and weather conditions.

The buoy is designed to withstand Sea State 7 conditions, which correspond to very rough ocean environments. However, predicting the exact environmental conditions that the buoy will encounter is inherently uncertain. Actual ocean conditions may be less severe than the assumed design conditions, or in rare cases may exceed the estimated values.

The environmental parameters used in ProteusDS were selected to represent conservative near-shore storm conditions along the Vancouver Island coastline. Because the buoy is deployed relatively close to shore, wave characteristics are influenced by bathymetry and coastal effects. These factors introduce uncertainty when estimating wave height and resulting mooring tension.

4.2 Modelling Adhesive Joint Interfaces Between Printed Sections

4.3 Effect of Infill Percentage and Pattern on Structural Performance

4.4 Hydrostatic Pressure Loading on Submerged Surfaces

A technical challenge associated with this loading condition is that the location of the waterline is not constant in real ocean environments. As waves pass the buoy, the instantaneous draft depth changes due to buoy motion and changing water surface elevation. This results in variations in the submerged surface area and the magnitude of hydrostatic pressure acting on the structure.

In this analysis, the hydrostatic pressure was applied using a constant draft depth corresponding to the static equilibrium position of the buoy. This represents an average loading condition and simplifies the simulation process. In reality, the buoy continuously moves in response to wave motion, causing the draft line to shift upward and downward over time.

Because the hydrostatic pressure varies dynamically in real conditions, the applied loading represents an approximation of the average pressure distribution. Despite this simplification, the loading condition provides a reasonable estimate of the structural stresses experienced by the buoy during operation.

4.5 Structural Optimization of Wall Thickness (PLA Model)

4.6 Structural Optimization of Wall Thickness (Polyethylene Model)

For the polyethylene (PE) buoy, the solid model was used. This best represents the physical buoy as the sliced model only holds true for the PLA buoy. First, the PE buoy was optimized as-is, with no additional features when compared to PLA buoy.

After the standard buoy, we repeated the optimization with a buoy with an additional rib in the high stress portion of the buoy, spanning the base to increase rigidity. To simplify the simulation, the following assumptions were made:

- There will be a single rib that spans the width of the buoy.
- The rib will be fixed at 1/2" in height to align with the extruded surface at the base of the buoy.
- The draft angle of the rib is neglected to reduce iterative operations.
- The width of the rib will be varied to optimize geometry.

The buoy with the rib can be found in **Figure 5**.

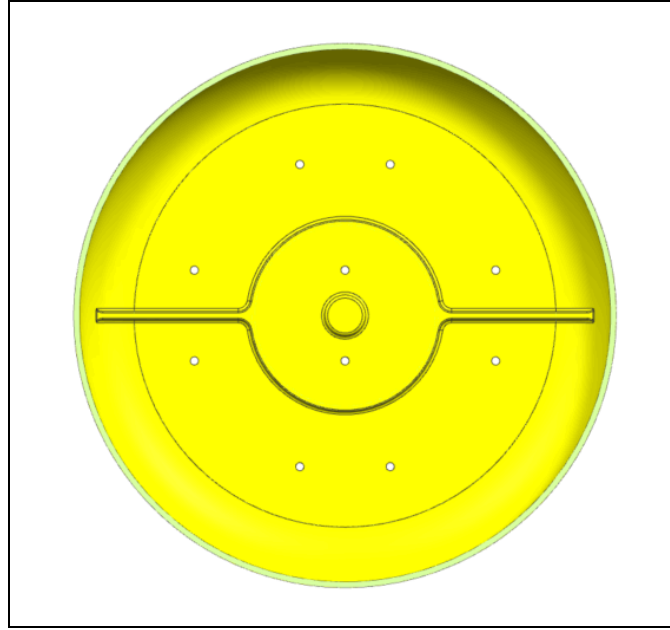


Figure 5: Section View of PE Buoy with Rib

Using an identical setup to that of the optimized buoy (wall thickness of $\sim 4.6\text{mm}$), we run an initial FEA stress solution to benchmark the addition of the rib. We observe an immediate reduction in stress from $\sim 10.5\text{MPa}$ to 9.59MPa as seen in figure 6.

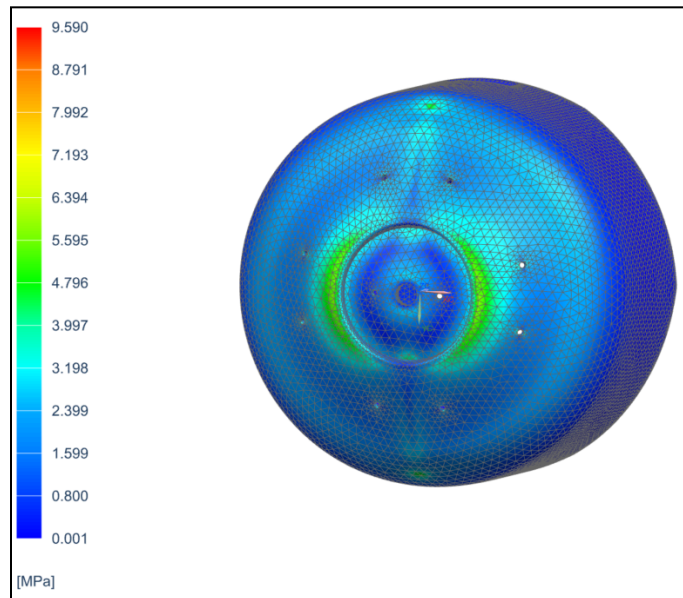


Figure 6: Unoptimized PE Buoy Rib Simulation

For this optimization solution, the design variables were set as rib thickness and wall thickness. See figure 7.

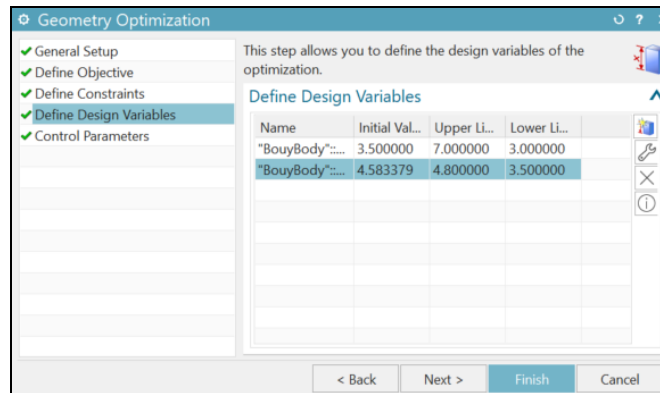


Figure 7: Rib Optimization Setup

Running this optimization yields a rib thickness of ~0.243in and a wall thickness of 4.52mm, as seen in figure 8.

Optimization History					
Based on Optimizer					
Design Objective Function Results					
Minimum Weight [N]	0	1	2	3	4
	37.04525	37.10118	35.00162	36.30261	36.52791
Design Variable Results					
Name	0	1	2	3	4
"BouyBody"::p1868=3.5	3.5	4.3	3.5	3.09017	3.09017
"BouyBody"::p355=4.583379	4.583379	4.583379	4.323379	4.492475	4.52115
Design Constraint Results					
Result Measure	0	1	2	3	4
Upper Limit = 10.500000 [MPa]	9.5902	9.0663	11.523	10.696	10.418
Small change in design, run converged.					

Figure 8: Rib Optimization Result

With an original weight of 36.8N, the difference in weight is negligible at an optimized weight of 36.5N. This converts to a ~30g difference, or less than 1% material savings, between ribbed and unribbed. The addition of a rib would also force the electronics to be supported by standoffs to prevent interference, which - based on design - would most likely negate much of this cost saving, along with increasing the cost through the newly required longer screws. If we saw a larger material saving this may have been a feasible route to investigate, but due to the small change, the non-ribbed buoy will be accepted as the final PE design.

5.0 Group Effort and Individual Contributions

Team Member	Contribution	Percentage
Luca	ProteusDS mooring simulations and research on the effect of PLA material properties on structural strength	33%
Rory	Finite Element Analysis of PLA segmented buoy and polyethylene buoy models	33%
Ronan	Structural optimization of wall thickness for PLA and polyethylene models	33%

6.0 Summary

6.1 Key findings

6.2 Structural feasibility of buoy design

6.3 Optimization outcomes

7.0 Experience and Suggestions

7.1 Experience using integrated CAD/CAE workflow

7.2 Limitations of modelling 3D printed materials

7.3 Recommendations for future work

7.4 Potential improvements to buoy structural design

References

Appendix A. CAD drawings of buoy geometry

Appendix B. Materials property tables for PLA

General information

Designation

PLA (general purpose), Polylactide / Polylactic acid (General purpose)

Tradenames

EcolGreen; ErcrosBio; Fozeeas; Futerro; Ingeo; Luminy; NaturePlast; Purasorb; Revode; Terramac

Typical uses

Biodegradable packaging and disposables, food packaging, plastic bags, consumer goods, diapers, bottles, cold drink cups, sheet, film products, electrical/electronic applications, automotive applications, home textiles, clothing, medical devices, homeware, personal care products, fibers, filaments, film, foam, laminates, trays.

Included in Materials Data for Simulation	✓
Materials Data for Simulation name	Plastic, PLA

Composition overview

Compositional summary

$(CH(CH_3)CO_2)_n$. The lactic acid is produced from sugar (dextrose) with plant starch origins e.g. corn, wheat, sugar beets and sugar cane.

Material family	Plastic (thermoplastic, semi-crystalline)
Base material	PLA (Polylactic acid / polylactide)
Renewable content	100 %
Polymer code	PLA

Composition detail (polymers and natural materials)

Polymer	100 %
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Price

Price	* 4.42	- 4.92	CAD/kg
Price per unit volume	* 5.3e3	- 6.35e3	CAD/m ³

Physical properties

Density	1.2e3	- 1.29e3	kg/m ³
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Mechanical properties

Young's modulus	2.4	- 3.6	GPa
Specific stiffness	1.92	- 2.9	MN.m/kg

Values marked * are estimates.
ANSYS, Inc. provides no warranty for this data.

Yield strength (elastic limit)	45	-	72	MPa
Tensile strength	45	-	72	MPa
Specific strength	36.1	-	58	kN.m/kg
Elongation	2.2	-	6	% strain
Elongation at yield	2.8	-	3.5	% strain
Compressive modulus	* 2.4	-	3.6	GPa
Compressive strength	* 54	-	86.4	MPa
Flexural modulus	2.6	-	4.6	GPa
Flexural strength (modulus of rupture)	90	-	126	MPa
Shear modulus	* 0.902	-	1.35	GPa
Shear strength	* 27	-	43.2	MPa
Bulk modulus	* 2.35	-	3.53	GPa
Poisson's ratio	0.33			
Shape factor	5			
Hardness - Vickers	* 16.3	-	17.9	HV
Hardness - Rockwell M	82	-	107	
Hardness - Rockwell R	* 32	-	35	
Hardness - Shore D	73	-	82	
Elastic stored energy (springs)	350	-	868	kJ/m ³
Fatigue strength at 10 ⁷ cycles	* 18	-	28.8	MPa

Impact & fracture properties

Fracture toughness	* 3.38	-	4.09	MPa.m ^{0.5}
Toughness (G)	* 3.62	-	6.12	kJ/m ²
Impact strength, notched 23 °C	1.5	-	5	kJ/m ²
Impact strength, notched -30 °C	1.6			kJ/m ²
Impact strength, unnotched 23 °C	13	-	23	kJ/m ²
Impact strength, unnotched -30 °C	25			kJ/m ²

Thermal properties

Melting point	135	-	181	°C
Glass temperature	43	-	63	°C
Heat deflection temperature 0.45MPa	51	-	60	°C
Heat deflection temperature 1.8MPa	48			°C
Vicat softening point	62.5			°C
Maximum service temperature	* 40	-	60	°C

Values marked * are estimates.
ANSYS, Inc. provides no warranty for this data.

Minimum service temperature	* -45	-	-12	°C
Thermal conductivity	* 0.12	-	0.15	W/m.°C
Specific heat capacity	* 1.18e3	-	1.21e3	J/kg.°C
Thermal expansion coefficient	* 68	-	78	µstrain/°C
Thermal shock resistance	* 196	-	360	°C
Thermal distortion resistance	* 0.00162	-	0.0021	MW/m

Electrical properties

Electrical resistivity	* 2.7e16	-	4.3e17	µhm.cm
Electrical conductivity	* 4.01e-16	-	6.39e-15	%ACS
Dielectric constant (relative permittivity)	* 3.1	-	3.2	
Dissipation factor (dielectric loss tangent)	* 0.00909	-	0.01	
Dielectric strength (dielectric breakdown)	* 3	-	6.2	MV/m

Magnetic properties

Magnetic type	Non-magnetic
---------------	--------------

Optical, aesthetic and acoustic properties

Refractive index	* 1.44	-	1.46	
Transparency	Transparent			
Acoustic velocity	1.38e3	-	1.71e3	m/s
Mechanical loss coefficient (tan delta)	* 0.0124	-	0.0164	

Critical materials risk

Contains >5wt% critical elements?	No
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Absorption & permeability

Water absorption @ 24 hrs	* 0.1	-	0.13	%
Water absorption @ sat	* 0.7	-	1	%
Humidity absorption @ sat	* 0.21	-	0.3	%
Water vapor transmission	9.5			g.mm/m^2.day
Permeability (O2)	16.9	-	17	cm^3.mm/m^2.day.atm

Processing properties

Polymer injection molding	Excellent
Polymer extrusion	Excellent
Polymer thermoforming	Excellent

Values marked * are estimates.
ANSYS, Inc. provides no warranty for this data.

Linear mold shrinkage	0.2	-	1.7	%
Melt temperature	180	-	221	°C
Mold temperature	30	-	120	°C
Molding pressure range	* 55.2	-	103	MPa

Durability

Water (fresh)	Acceptable			
Water (salt)	Acceptable			
Weak acids	Acceptable			
Strong acids	Unacceptable			
Weak alkalis	Acceptable			
Strong alkalis	Unacceptable			
Organic solvents	Limited use			
Oxidation at 500C	Unacceptable			
UV radiation (sunlight)	Good			
Flammability	Highly flammable			
Oxygen index	* 19	-	21	%

Environmental indicators - primary production

Climate change (CO2-eq), primary production (virgin grade)	* 2.72	-	2.99	kg/kg
* ecoinvent v3.10, market for polylactic acid, granulate/(GLO); Institute for Prospective Technological Studies, 2005; Vnik et al. 2007				
Climate change (CO2-eq), primary production (typical grade)	* 2.72	-	2.99	kg/kg
Embodied energy, primary production (virgin grade)	* 51.4	-	56.6	MJ/kg
* ecoinvent v3.10, market for polylactic acid, granulate/(GLO); Institute for Prospective Technological Studies, 2005; Vnik et al. 2007				
Embodied energy, primary production (typical grade)	* 51.4	-	56.6	MJ/kg
Water usage	* 19.8	-	21.8	l/kg

Environmental indicators - processing

Polymer extrusion energy	* 5.65	-	6.23	MJ/kg
Polymer extrusion CO2	* 0.424	-	0.467	kg/kg
Polymer extrusion water	* 4.75	-	7.13	l/kg
Polymer molding energy	* 13.5	-	14.9	MJ/kg
Polymer molding CO2	* 1.01	-	1.12	kg/kg
Polymer molding water	* 10.5	-	15.7	l/kg
Coarse machining energy (per unit wt removed)	* 0.969	-	1.07	MJ/kg
Coarse machining CO2 (per unit wt removed)				

Values marked * are estimates.
ANSYS, Inc. provides no warranty for this data.

	* 0.0727	-	0.0802	kg/kg
Fine machining energy (per unit wt removed)	* 5.41	-	5.96	MJ/kg
Fine machining CO2 (per unit wt removed)	* 0.406	-	0.447	kg/kg
Grinding energy (per unit wt removed)	* 10.3	-	11.4	MJ/kg
Grinding CO2 (per unit wt removed)	* 0.774	-	0.853	kg/kg

Recycling and end of life

Recycle	✗			
Functional recycle	✗			
Combust for energy recovery	✓			
Heat of combustion (net)	* 18.9	-	19.9	MJ/kg
Combustion CO2	* 1.8	-	1.9	kg/kg
Landfill	✓			
Biodegrade	✓			

Notes

Other notes

PLA is a renewable thermoplastic polyester manufactured from plants such as sugarcane, corn and tapioca. PLA can be amorphous or semi-crystalline. Various blends of D and L enantiomers are available, making available a broader range of properties.

Links

[ProcessUniverse](#)

[Producers](#)

[Reference](#)

[Shape](#)